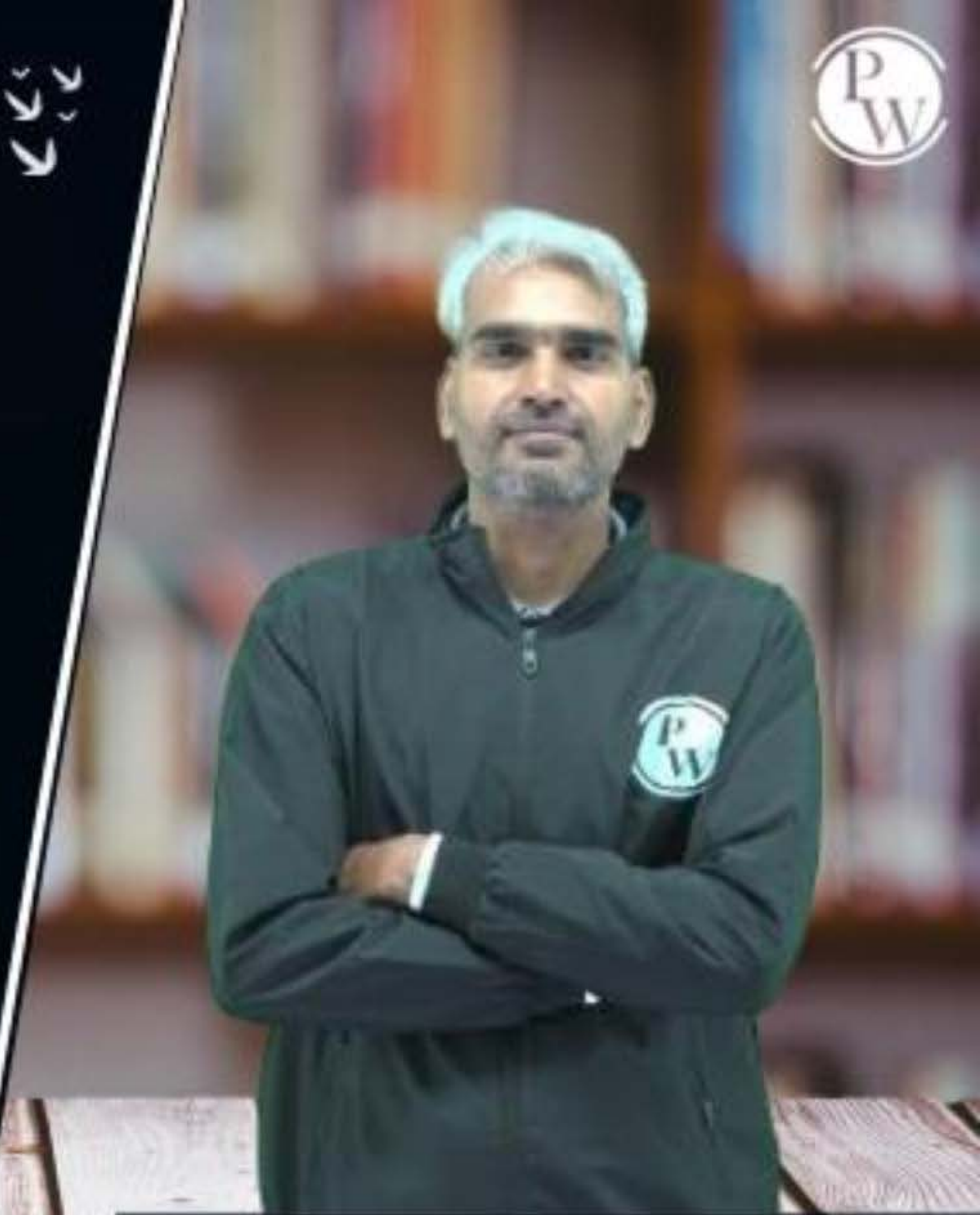
Computer Science & IT

Database Management System

Query Languages

Lecture No. 04



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Recap of Previous Lecture

- **Topic** Division operation
- **Topic** Practice questions

Slide

Topics to be Covered

Topic Practice questions

Slide

Topic : Division (÷)

Division operation is used whenever the query is with respect to every or all.

Student (S)

Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Enroll (E)

Sid	Cid	- - - - ..
S_1	C_1	
S_1	C_2	
S_2	C_2	
S_1	C_3	
S_2	C_3	
S_3	C_3	

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

Query:- Retrieve Sids of the students who have enrolled for all Courses.

Enroll (E)

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_1	C_3
S_2	C_3
S_3	C_3

$$\underbrace{\pi_{sid}(E)}_{} \Rightarrow \quad o/p =$$

Sid
S_1
S_2
S_3

This R.A. Expression will produce the Sids of the Students who have enroll for some Course {at least}

Course(C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

$$\underbrace{\pi_{cid}\ (C)}_{\uparrow} \implies \text{o/p} =$$

Cid
C_1
C_2
C_3

This R.A. Expression will produce Cids of all courses.

* $\pi_{sid}(E) =$

Sid
S_1
S_2
S_3

* $\pi_{cid}(C) =$

Cid
C_1
C_2
C_3

$\underbrace{\pi_{sid}(E) \times \pi_{cid}(C)}$ $\Rightarrow$ o/p =

E.Sid	C.Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_3

This R.A. Expression will produce a universal relation, in which Every student who has Enrolled for some Courses is Combined with all Courses.

Student (S)

Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Enroll (E)

Sid	Cid	fee
S_1	C_1	
S_1	C_2	
S_2	C_2	
S_1	C_3	
S_2	C_3	
S_3	C_3	

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

* Retrieve Sids of the students who have Enrolled for all Courses.

* We are actually looking for the Sids in the Enroll table which are associated with all the Cids of the Course table

* W.r.t. given relational instance O/p Should be $\boxed{S_1}$ because it is the Only Sid in Enroll table which is associated with all the Cids of Course table.

W.r.t. "all" we can use division operation,

i.e., $\pi_{sid,cid}(E) \div \pi_{cid}(C)$

Student (S)

Sid	Sname
S_1	A
S_2	A
S_3	B
S_4	C

Enroll (E)

Sid	Cid	fee
S_1	C_1	
S_1	C_2	
S_2	C_2	
S_1	C_3	
S_2	C_3	
S_3	C_3	

Course (C)

Cid	Cname
C_1	OS
C_2	COA
C_3	DBMS

* Derivation of division (÷) operation using basic relational algebra operation

$$\pi_{sid,cid}(E) \div \pi_{cid}(C)$$

$$\equiv \pi_{sid}(E) - \pi_{sid}\left[\underbrace{\pi_{sid}(E) \times \pi_{cid}(C)} - \pi_{sid,cid}(E)\right]$$

$\pi_{sid}(E)$:

Sid
S_1
S_2
S_3

Sids of the students who have enroll for some courses

Result:

Sid
S_1

Sids of the students who have enrolled for all courses

$\pi_{sid}[\ldots]$:

Sid
S_2
S_3

In the o/p we will get the Sids of the students who did not enroll for all courses

$\pi_{sid}(E) \times \pi_{cid}(C)$:

Sid	Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_2

Result of the subtraction:

Sid	Cid
S_2	C_1
S_3	C_1
S_3	C_2

In the output of this subtraction we will get the Sids of the students who did not enroll for all courses along with the Cids of the course for which they did not enroll.

$\pi_{sid,cid}(E)$:

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_1	C_3
S_2	C_3
S_3	C_3

$$\pi_{sid,cid}(E) \div \pi_{cid}(C)$$

$$\equiv \underset{1}{\pi_{sid}(E)} \underset{8}{-} \underset{7}{\pi_{sid}} \left[\underset{2}{\pi_{sid}(E)} \underset{4}{\times} \underset{3}{\pi_{cid}(C)} \underset{6}{-} \underset{5}{\pi_{sid,cid}(E)} \right]$$

Order of Execution

- Consider the following relational tables:
  - Supplier (<u>Sid</u>, Sname, Rating)
  - Parts (<u>Pid</u>, Pname, Color)
  - Catalog (<u>Sid</u>, <u>Pid</u>, Cost)

Supplier

Sid	Sname	Rating
S_1	A	3
S_2	A	5
S_3	B	7
S_4	C	0

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Q:- Retrieve Sids of all the suppliers.

$\pi_{sid}(\text{Supplier})$

O/P :

Sid
S_1
S_2
S_3
S_4

Q:- Retrieve Sids of Suppliers who have supplied at least one part.

$\pi_{sid}(\text{Catalog})$

O/P =

Sid
S_1
S_2
S_3

#Q. Retrieve Sid of the suppliers whose rating is more than 5.

$$\pi_{sid}\left(\underbrace{\sigma_{Rating > 5}(Supplier)}\right)$$

sid	Sname	Rating
S_3	B	7

o/p = $\boxed{S_3}$

#Q. Retrieve Sid of the suppliers who have supplied some parts.

$\underbrace{\text{some}}_{\text{i.e. (at least one)}}$

$\pi_{sid}(\text{Catalog}) \Rightarrow$ o/p :-

Sid
S_1
S_2
S_3

#Q. Retrieve Sid of the suppliers who have supplied some Red color parts.

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

<u>Pid</u>	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

(i) $\pi_{Catalog.Sid}\left[\sigma_{Catalog.Pid = Parts.Pid \wedge Parts.Color = 'Red'}(Catalog \times Parts)\right]$

O/P:

Sid
S_1
S_3

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

(ii) $\pi_{Sid} \left[\sigma_{Color='Red'} \left(Catalog \bowtie Parts \right) \right]$

$\downarrow$

O/P =

S_1
S_3

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

Catalog

<u>Sid</u>	<u>Pid</u>	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

<u>Pid</u>	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

(iii) $\pi_{\text{Catalog.Sid}}\left(\sigma_{\text{catalog.Pid = Parts.Pid}}\left(\text{Catalog} \times \left(\sigma_{\text{Color='Red'}}(\text{Parts})\right)\right)\right)$

(iv) $\pi_{\text{sid}}\left(\text{Catalog} \bowtie \left(\sigma_{\text{Color='Red'}}(\text{Parts})\right)\right)$

These two query are more efficient Compared to previous two queries.

$$\left(\rho_C(\text{Catalog}) \times \rho_P(\text{Parts})\right)$$

(i) $\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid \wedge P.Color = 'Red'}(C \times P)\right)$

(ii) $\pi_{Sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right)$

$\Big\}$ less efficient

(iii) $\pi_{C.Sid}\left(\sigma_{C.Pid=P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right)$

(iv) $\pi_{Sid}\left(C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right)$

$\Big\}$ More efficient

#Q. Retrieve Sid of the suppliers who have supplied some Red or some Green color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

C×P

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

(i) $$\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid \,\wedge\, (P.Color = \text{'Red'} \,\vee\, P.Color = \text{'Green'})}(C \times P)\right)$$

(ii) $$\pi_{Sid}\left(\sigma_{(color = \text{'Red'} \,\vee\, color = \text{'Green'})}(C \bowtie P)\right)$$

(iii) $$\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color = \text{'Red'} \,\vee\, Color = \text{'Green'}}(P)\right)\right)\right)$$

(iv) $$\pi_{Sid}\left(C \bowtie \left(\sigma_{Color = \text{'Red'} \,\vee\, Color = \text{'Green'}}(P)\right)\right)$$

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

CXP

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

(i) $\pi_{C.Sid}\left(\sigma_{\substack{C.Pid = P.Pid \\ \wedge \\ (P.Color = 'Red' \wedge P.Color = 'Green')}}(C \times P)\right)$

(ii) $\pi_{Sid}\left(\sigma_{(Color = 'Red' \wedge Color = 'Green')}(C \bowtie P)\right)$

(iii) $\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color = 'Red' \wedge Color = 'Green'}(P)\right)\right)\right)$

(iv) $\pi_{Sid}\left(C \bowtie \left(\sigma_{Color = 'Red' \wedge Color = 'Green'}(P)\right)\right)$

All are Wrong query to get the desired result.

$\Downarrow$

In a single tuple of "C×P", Color Can never be "Red" as well as "green" at the same time

$\therefore$ All of them will Produce Empty output

(i) $\pi_{C.Sid}\left(\sigma_{\substack{C.Pid = P.Pid \\ \wedge \\ P.Color = 'Red'}}(C \times P)\right) \cap \pi_{C_1.Sid}\left(\sigma_{\substack{C_1.Pid = P_1.Pid \\ \wedge \\ P_1.Color = 'Green'}}(C_1 \times P_1)\right)$

(ii) $\pi_{Sid}\left(\sigma_{Color='Red'}(C \bowtie P)\right) \cap \pi_{Sid}\left(\sigma_{Color='Green'}(C_1 \bowtie P_1)\right)$

(iii) $\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color='Red'}(P)\right)\right)\right) \cap \pi_{C_1.Sid}\left(\sigma_{C_1.Pid = P_1.Pid}\left(C_1 \times \left(\sigma_{Color='Green'}(P_1)\right)\right)\right)$

(iv) $\pi_{Sid}\left(C \bowtie \left(\sigma_{Color='Red'}(P)\right)\right) \cap \pi_{Sid}\left(C_1 \bowtie \left(\sigma_{Color='Green'}(P_1)\right)\right)$

Catalog

Sid	Pid	Cost
S_1	P_1	10
S_2	P_2	20
S_3	P_3	30

Parts

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	XYZ	Red

$C_1 \times P_1$

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

$C_2 \times P_2$

C.Sid	C.Pid	C.Cost	P.Pid	P.name	P.Color
S_1	P_1	10	P_1	XYZ	Red
S_1	P_1	10	P_2	PQR	Green
S_1	P_1	10	P_3	XYZ	Red
S_2	P_2	20	P_1	XYZ	Red
S_2	P_2	20	P_2	PQR	Green
S_2	P_2	20	P_3	XYZ	Red
S_3	P_3	30	P_1	XYZ	Red
S_3	P_3	30	P_2	PQR	Green
S_3	P_3	30	P_3	XYZ	Red

$$= \pi_{C_1.Sid}\left(\sigma_{\left((C_1.Sid = C_2.Sid) \wedge (C_1.Pid = P_1.Pid \wedge P_1.Color = 'Red') \wedge (C_2.Pid = P_2.Pid \wedge P_2.Color = 'green') \right)} \left(\underbrace{(C_1 \times P_1)} \times \underbrace{(C_2 \times P_2)} \right) \right)$$

let us use $C_1 \times P_1$ to select the tuple w.r.t. 'Red' Color part

and

let us use $C_2 \times P_2$ to select the tuple w.r.t. 'green' Color part

we can also use $C_2.Sid$ as well because both of them are same

Q:- Retrieve Sids of suppliers who have supplied some "red" or some "green" Color Parts

(i) $\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid \wedge P.Color = 'Red'}(C \times P)\right) \cup \pi_{C_1.Sid}\left(\sigma_{C_1.Pid = P_1.Pid \wedge P_1.Color = 'Green'}(C_1 \times P_1)\right)$

(ii) $\pi_{Sid}\left(\sigma_{Color = 'Red'}(C \bowtie P)\right) \cup \pi_{Sid}\left(\sigma_{Color = 'Green'}(C_1 \bowtie P_1)\right)$

(iii) $\pi_{C.Sid}\left(\sigma_{C.Pid = P.Pid}\left(C \times \left(\sigma_{Color = 'Red'}(P)\right)\right)\right) \cup \pi_{C_1.Sid}\left(\sigma_{C_1.Pid = P_1.Pid}\left(C_1 \times \left(\sigma_{Color = 'Green'}(P_1)\right)\right)\right)$

(iv) $\pi_{Sid}\left(C \bowtie \left(\sigma_{Color = 'Red'}(P)\right)\right) \cup \pi_{Sid}\left(C_1 \bowtie \left(\sigma_{Color = 'Green'}(P_1)\right)\right)$

Q: Retrieve Sids of the suppliers who have supplied some "red" or some "green" color parts

$$= \pi_{C_1.Sid}\left(\sigma_{\left((C_1.Sid = C_2.Sid) \wedge \left((C_1.Pid = P_1.Pid \wedge P_1.Color = 'Red') \vee (C_2.Pid = P_2.Pid \wedge P_2.Color = 'green')\right)\right)}\left(\underbrace{(C_1 \times P_1)} \times \underbrace{(C_2 \times P_2)}\right)\right)$$

let us use $C_1 \times P_1$ to select the tuple w.r.t. 'Red' color part

and

let us use $C_2 \times P_2$ to select the tuple w.r.t. 'green' color part

we can also use $C_2.Sid$ as well because both of them are same

#Q. Retrieve Sid of the suppliers who have supplied all parts.

Catalog (Sid, Pid, Cost)

Parts (Pid, Pname, Color)

We are interested in the Sids in the catalog table Which are associated With all Pids of Parts table

$\therefore$ Query will be

$$\pi_{Sid, Pid}(Catalog) \div \pi_{Pid}(Parts)$$

$$= \left[\pi_{Sid}(C) - \pi_{Sid}\left[\left(\pi_{Sid}(C) \times \pi_{Pid}(P)\right) - \pi_{Sid, Pid}(C)\right]\right]$$

#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

We are actually looking for Sids in Catalog table which are associated with all Pids of "Red" Color Parts.

$$\pi_{Sid,Pid}(C) \div \pi_{Pid}\left(\sigma_{Color='Red'}(P)\right)$$

$$= \left[\pi_{Sid}(C) - \pi_{Sid}\left[\left(\pi_{Sid}(C) \times \pi_{Pid}\left(\sigma_{Color='Red'}(P)\right)\right) - \pi_{Sid,Pid}(C)\right]\right]$$

#Q. Retrieve Sid of the suppliers who have supplied at least two parts.

* For the upcoming questions we will consider the following Catalog relation.

Catalog

Sid	Pid	Cost
S_1	P_1	20
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	10

Q. Retrieve Sids of the suppliers who have supplied at least two Parts.

Catalog (C_1)

Sid	Pid	Cost
S_1	P_1	20
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	10

Catalog (C_2)

Sid	Pid	Cost
S_1	P_1	20
S_1	P_2	30
S_2	P_2	30
S_3	P_2	20
S_3	P_3	10

We will have to compare the Sid of a tuple of Catalog table with Sids of all other tuples of the catalog table.

If the same Sid is present in any other tuple of Catalog table, that supplier must have supplied at least two Parts.

$$= \pi_{C_1.Sid}\left[\sigma_{((C_1.Sid = C_2.Sid) \wedge (C_1.Pid\ != C_2.Pid))}(C_1 \times C_2)\right]$$

If we do not use this Cond^n then each Sid from catalog will be present in o/p

Because each tuple will Combine with itself

#Q. Retrieve Sid of the suppliers who have supplied exactly one part.

Suppliers who have supplied Exactly one Part = (Suppliers who have supplied at least one part) − (Suppliers who have supplied at least two parts)

1 or 2 or 3 or more parts

2 or 3 or 4 or more parts

$$= \pi_{Sid}(\text{Catalog}) - \left\{ \pi_{C_1.Sid}\left(\sigma_{(C_1.Sid = C_2.Sid \wedge C_1.Pid <> C_2.Pid)} (C_1 \times C_2) \right) \right\}$$

Not Equal

#Q. Retrieve Sid of the suppliers who have supplied $\underbrace{\text{at most one part}}^{\text{i.e., "0 or 1"}}$.

= All suppliers − Suppliers who have supplied at least two parts

{0 or more parts} − {2 or more parts}

$$= \pi_{Sid}(\text{Supplier}) - \pi_{C_1.Sid}\left(\sigma_{(C_1.Sid = C_2.Sid) \wedge (C_1.Pid <> C_2.Pid)}(C_1 \times C_2)\right)$$

* H.W.

#Q. Retrieve Sid of the suppliers who have supplied $\underbrace{\text{at least three parts}}_{(C_1 \times C_2 \times C_3)}$.

H.W.

#Q. Retrieve Sid of the suppliers who have supplied most expensive parts.

Hint:

π_{Sid} [(All tuples from catalog) − (Tuples of catalog table in which cost is not maximum)]

Tuples from catalog in which cost is maximum.

2 mins Summary

Topic Practice questions

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THANK - YOU

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